

ABOUT ME

Results-driven Data Engineer with over 6+ years of experience designing and implementing scalable data architectures across cloud platforms including AWS, Azure, and GCP. Skilled at transforming raw, high-volume datasets into optimized pipelines and analytical solutions that enable data-driven decision-making and business growth.

CERTIFICATIONS



TECHNICAL SKILLS

Programming & Scripting: Python, Java, Scala, SQL, Bash.

Big Data Technologies: Hadoop, Apache Spark, Hive, HDFS, Presto, Pig, Flink

ETL/ELT Tools: Informatica, Talend, Apache Nifi, dbt, SSIS

Data Warehousing: Snowflake, Redshift, Azure Synapse, Big Query, Teradata, Vertica.

Data Integration & Orchestration: Apache Airflow, Luigi, AWS Step Functions, Azure Logic Apps, GCP Data Fusion

Streaming & Real-Time Processing: Apache Kafka, AWS Kinesis, Azure Event Hubs, Google Pub/Sub, Spark Streaming.

Cloud Platforms:

AWS: Redshift, S3, Glue, Lambda, EMR, Athena, DynamoDB, Kinesis, CloudFormation

Azure: Azure Data Factory, Synapse Analytics, Azure Storage, Databricks, Cosmos DB, Logic Apps, Azure Functions

Database Systems: PostgreSQL, MySQL, Oracle, SQL Server, MongoDB, Cassandra, DynamoDB, Cosmos DB, HBase

Business Intelligence Tools: Tableau, Power BI, AWS Quick Sight, Google Data Studio

DevOps & CI/CD: Jenkins, GitLab CI/CD, Azure DevOps, AWS Code Pipeline, Docker, Kubernetes

Machine Learning Integration: AWS Sage Maker, Azure ML, GCP Vertex AI

Workflow Automation: Terraform, CloudFormation,

EDUCATION

- Master's in computer and information sciences
Western Illinois University

PROFISSIONAL SUMMARY

- Seasoned Data Engineer with over **6+** years of experience in architecting and delivering large-scale data platforms, transforming unstructured data into structured formats that fuel analytics, machine learning, and operational reporting.
- Deep expertise in building robust ETL pipelines and batch workflows using **Apache Airflow**, **Jenkins**, and **Azure Data Factory**, with strong adherence to CI/CD principles for automated and version-controlled deployments.
- Proficient in cloud-native data services across **AWS**, **Azure**, and **GCP**, including hands-on work with **AWS Glue**, **Redshift**, **S3**, **Lambda**, **Azure Synapse Analytics**, **Dataproc**, and **BigQuery** to develop scalable solutions.
- Strong command over big data tools like **Hadoop**, **Hive**, **Spark**, **Pig**, and **MapReduce**, leveraging distributed computing to process and analyze large datasets efficiently across multi-node clusters.
- Solid programming background in **Python**, **Java**, and **Scala** for building data ingestion frameworks, developing transformation scripts, and integrating custom business logic into data pipelines.
- Experienced in designing real-time data processing pipelines using **Apache Kafka**, **Amazon Kinesis**, and **Spark Streaming**, enabling near real-time analytics and event-driven data architectures.
- Skilled in working with both **SQL** and **NoSQL** databases such as **PostgreSQL**, **Oracle**, **MySQL**, **MongoDB**, and **Cassandra**, focusing on schema optimization, indexing, and performance tuning.
- Well-versed in data warehousing and data lake design principles, applying techniques like partitioning, schema evolution, and query optimization to improve scalability and performance.
- Applied industry-standard compliance and governance practices including **GDPR**, **HIPAA**, and **CCPA**, securing sensitive data through access controls, audit trails, and encryption strategies.
- Delivered end-to-end data integration across hybrid systems using **Sqoop**, **Flume**, **Zookeeper**, and **Impala** in collaboration with cross-functional teams.

WORK EXPERIENCE

AWS Data Engineer || Comerica Incorporated, Dallas, Texas || May 2025 - Present

Description: Comerica Incorporated is an American financial services company. Designed and deployed robust ETL pipelines to transform and standardize raw data into structured, analytics-ready formats, significantly enhancing the efficiency of reporting and business intelligence workflows. Established comprehensive monitoring and logging frameworks to proactively detect anomalies, ensure high pipeline reliability, and uphold data integrity throughout the end-to-end data lifecycle.

Responsibilities

- Developed and maintained scalable ETL/ELT pipelines using **AWS Glue**, **Apache Spark**, and **AWS Lambda** to process high-volume transactional and customer data for real-time fraud detection and regulatory compliance.
- Architected cloud-native data platforms leveraging **Amazon S3**, **Amazon Redshift**, **Athena**, and **DynamoDB**, centralizing CRM, banking, and risk system data to power customer profiling and credit risk dashboards.
- Executed large-scale batch processing using **Apache Spark**, **Amazon EMR**, and **Hadoop**, enabling support for anti-money laundering (AML) efforts and predictive credit scoring models.
- Designed real-time data ingestion pipelines with **Amazon Kinesis**, **AWS Lambda**, and **Apache Spark Streaming**, enabling near real-time tracking of customer activity and fraud indicators.
- Built automated monitoring and alerting systems using **Amazon CloudWatch**, **Prometheus**, and **Grafana**, ensuring operational stability and proactive issue resolution across all streaming and batch pipelines.
- Deployed CI/CD workflows using **AWS CodePipeline**, **CodeBuild**, and **CodeDeploy**, reducing deployment risks and enabling seamless analytics infrastructure updates.
- Automated infrastructure provisioning using **AWS CloudFormation** and **Terraform**, creating secure, scalable environments for financial data analytics and regulatory workloads.
- Leveraged **AWS Glue DataBrew**, **Athena**, and **SQL** to clean, validate, and transform datasets, improving data quality and ensuring audit readiness for compliance with **Basel III** and **IFRS 9**.
- Engineered serverless data workflows with **AWS Lambda** to process streaming logs and fraud signals, supporting analytics use cases such as customer segmentation and risk forecasting.
- Collaborated with data scientists to operationalize machine learning models via **Amazon SageMaker**, with integration into production systems using **S3**, **Lambda**, and custom APIs.
- Applied robust data governance using **AWS Glue Catalog**, **Lake Formation**, and **IAM**, enabling secure data access and streamlined metadata management across multiple business units.
- Migrated legacy **SQL Server** and **Oracle** systems to **Amazon Redshift** and **Amazon Aurora** using **AWS DMS**, ensuring minimal downtime and seamless data continuity.
- Optimized data lake performance on **Amazon S3** through advanced partitioning, compression, and lifecycle rules, leading to improved query efficiency and reduced storage costs.
- Created reusable ingestion frameworks using **Python**, **PySpark**, and **SQL**, standardizing ETL logic across disparate source systems and improving onboarding velocity for new data feeds.
- Integrated data pipelines with visualization platforms like **Power BI**, **Tableau**, and **Amazon QuickSight**, enabling near real-time business KPI reporting for decision-makers.

Environment: AWS Glue, Apache Spark, AWS Lambda, Amazon S3, Amazon Redshift, Amazon Athena, Amazon EMR, Amazon Kinesis, AWS CloudFormation, Terraform, Python, IAM, AWS Lake Formation, Power BI, Tableau.

Azure Data Engineer || Medtronic, Minneapolis, Minnesota || May 2024 - April 2025

Description: Medtronic plc is an American-Irish medical device company. Designed and implemented end-to-end data integration pipelines on the Azure cloud to support scalable data warehousing and lakehouse solutions. Streamlined ingestion and inter-team data sharing by leveraging services like Azure SQL Database and Azure Data Lake, ensuring secure, well-structured, and easily accessible data assets for analytics and reporting across the organization.

Responsibilities

- Developed and orchestrated robust ETL/ELT workflows using **Azure Data Factory (ADF)** to process and load complex financial and transactional datasets into **Azure Data Lake Storage (ADLS)** and **Azure Synapse Analytics** for enterprise reporting and compliance.

- Designed cloud-native data warehouse architectures using **Azure Synapse**, **Azure SQL Database**, and **SQL Server**, enabling real-time analytics for credit risk modeling, financial forecasting, and operational reporting.
- Leveraged **Azure Blob Storage**, **Data Lake Gen2**, and **Azure Storage Accounts** for secure handling of structured and semi-structured data, supporting payment processing and remittance operations at scale.
- Deployed and integrated ML models using **Azure Machine Learning**, **MLflow**, and **Synapse pipelines**, embedding predictive outputs into enterprise systems for fraud detection and customer churn analysis.
- Automated end-to-end deployment workflows with **Azure DevOps**, **Git**, and **ARM templates**, streamlining CI/CD processes, version control, and environment provisioning for data engineering projects.
- Designed interactive dashboards and real-time KPIs using **Power BI**, sourcing data from **Azure Analysis Services**, **Synapse**, and **SQL Server** to drive data-backed decisions across operations and finance.
- Built scalable microservices for fraud analytics by containerizing data services with **Docker** and deploying them via **Azure Kubernetes Service (AKS)** in a secure and distributed environment.
- Engineered data transformation and cleansing routines using **Python**, **SQL**, and **Scala**, ensuring data quality across analytics pipelines, and processed large-scale behavioural data using **Hadoop**, **Hive**, and **Cosmos DB**.
- Implemented real-time and batch streaming pipelines using **Apache Spark**, **Databricks**, and **Scala**, delivering rapid insights into customer behaviour, transaction trends, and remittance metrics.
- Integrated hybrid cloud solutions using **Snowflake**, **SnowSQL**, **Snowflake Streams**, **Azure Data Factory**, and **Azure Databricks** to deliver near real-time reporting for critical business indicators.
- Enhanced data observability and pipeline reliability by configuring real-time alerts and logs with **Azure Monitor**, **Log Analytics**, and **Application Insights**, addressing failures before impacting downstream systems.
- Enforced enterprise-grade data governance using **Azure Purview**, **Azure RBAC**, **Key Vault**, and **Managed Identities**, aligning data usage with **GDPR**, **SOC 2**, and internal audit requirements.
- Improved data quality and validation using frameworks like **Great Expectations** and **PyDeequ**, enabling automated integrity checks for incoming transactional and remittance data before further processing.

Environment: Azure Data Factory, Azure Data Lake Storage (ADLS), Azure Synapse Analytics, Azure SQL Database, Azure Machine Learning, Power BI, Azure DevOps, Git, Azure Kubernetes Service (AKS), Docker, Python, Scala, Hadoop, Hive, Pig, Impala, Azure Event Hubs, Azure Key Vault, Azure Functions, Apache Kafka, Apache Airflow, Terraform.

GCP Data Engineer || McKesson Corporation, Irving, Texas || Sep 2022 - April 2024

Description: McKesson Corporation is an American publicly traded company that distributes pharmaceuticals and provides health information technology, medical supplies, and health management tools. Improved data pipeline efficiency by optimizing system performance, automating data workflows, and integrating ML outputs, resulting in scalable and reliable analytics-ready data across platforms.

Responsibilities

- Developed scalable, high-throughput data pipelines using **Google Cloud Dataflow**, **BigQuery**, and **Cloud Data Fusion** to process large volumes of insurance policy and claims data for near real-time forecasting.
- Engineered robust ETL workflows to ingest and harmonize data from **Cloud SQL**, **Cloud Storage**, and third-party APIs, enabling efficient operational insights.
- Created interactive **Power BI** dashboards integrated with **BigQuery** to empower business teams with real-time analytics on claims, policy renewals, and customer engagement.
- Automated infrastructure provisioning and CI/CD pipelines using **Terraform**, **Cloud Build**, and **GitOps**, enhancing deployment speed and system reliability.
- Managed distributed data processing using **Hadoop**, **Apache Spark**, and **Hive** to transform semi-structured underwriting and fraud detection datasets.
- Developed custom data transformation logic with **Python** and **SQL** supporting advanced analytics and machine learning models for risk assessment and fraud scoring.
- Tuned query performance in **Hive** and **Impala**, reducing execution times for actuarial dashboards and regulatory reports.
- Maintained metadata, schema evolution, and data governance using **GCP Data Catalog** and **BigQuery**, improving data discoverability and lineage tracking.

- Implemented enterprise-grade security controls with **GCP IAM**, **Cloud KMS**, and **VPC Service Controls** to safeguard sensitive insurance data and ensure compliance.
- Orchestrated streaming data pipelines leveraging **Cloud Pub/Sub**, **Cloud Functions**, and **BigQuery** to enable near real-time fraud detection alerts.
- Monitored cloud costs and usage through **GCP Billing**, **BigQuery Audit Logs**, and custom dashboards in **Looker Studio** to optimize resource consumption and budget planning.

Environment: BigQuery, Dataflow, Cloud Data Fusion, Cloud Storage, Cloud SQL, Power BI, Terraform, Cloud Build, Git, Hadoop, Apache Spark, Hive, Python, SQL, Sqoop, Zookeeper, Impala, Pig, HDFS, AWS S3, Redshift, AWS Glue, GCP Data Catalog, GCP IAM, Cloud KMS, VPC Service Controls

Sr. Data Engineer || Aegon Life Insurance, Hyderabad, India || Aug 2020 - Nov 2021

Description: Aegon Ltd. is a Dutch public company for life insurance, pensions and asset management. I have played a key role in designing and maintaining secure, compliant data infrastructure, developing efficient pipelines to gather and analyze vehicle data that drives accurate maintenance forecasting.

Responsibilities

- Utilized **Azure Data Factory** to develop and automate pipelines that extract data from diverse sources and load it into **Azure SQL Database**. Scheduled these workflows using triggers and implemented temporary views to organize curated datasets efficiently.
- Built data pipelines that ingested telemetry data, transforming it into structured formats suitable for reporting. Leveraged **Databricks Notebooks** with **PySpark** and **SQL** to cleanse and prepare data for business analysts.
- Played a key role in migrating data from legacy systems to **Azure SQL Database** while optimizing and modifying existing **SQL Views** to improve reporting capabilities and generate business insights.
- Designed and maintained entry-level **SSIS** packages for ETL workflows within the data warehouse, using tools such as **BIDS**, **SSIS**, and **SSMS** to ensure accurate source-to-target data mappings and query validations.
- Executed data profiling and filtering on large datasets stored in **HDFS** by writing complex **HiveQL** queries. Employed **Impala** to conduct low-latency interactive queries for timely data retrieval.
- Managed the import of relational data into **Hive** tables with **Sqoop**, performing validation checks and troubleshooting to maintain data integrity throughout the Hadoop environment.
- Ensured data consistency and quality across **HDFS** layers by applying best practices in storage and processing, facilitating reliable downstream data consumption.

Environment: Azure Data Factory (ADF), Azure SQL Database, Azure Data Explorer/Kusto, Databricks, PySpark, SQL, SSIS, SSMS, BIDS, Hive, Hadoop, HDFS, Impala, Sqoop.

Data Engineer || Exide Life Insurance, Hyderabad, India || June 2019 - July 2020

Description: Exide Life Insurance Company Limited was an Indian life insurance company. Supported the improvement of data pipelines by optimizing performance and automating key workflows. Assisted in integrating machine learning outputs to help deliver accurate and scalable analytics-ready data.

Responsibilities

- Assisted in developing and scheduling **Data Factory** pipelines to extract and load data from various sources into **SQL Database**, working under senior guidance to ensure smooth daily data operations.
- Supported the ingestion and transformation of telemetry data into structured formats by contributing to data pipeline development using **Data Factory** and performing transformations in **Databricks Notebooks** with **PySpark** and **SQL**.
- Helped maintain and update existing **SQL Views** to support reporting needs, collaborating with analysts to improve data accessibility and accuracy for business insights.
- Contributed to building and validating **SSIS** packages for ETL workflows using **BIDS**, **SSIS**, and **SSMS**, assisting in source-to-target data mapping and query testing within the data warehouse environment.
- Performed data imports into **Hive** tables using **Sqoop** and participated in data validation and troubleshooting activities within the **Hadoop** ecosystem, including **Hive**, **HDFS**, and **Impala**, to ensure data integrity.
- Followed best practices to maintain data quality and consistency within **HDFS**, and actively took part in **Agile Scrum** meetings such as sprint planning and retrospectives to support team collaboration and delivery.

Environment: Databricks, PySpark, SQL, SSIS, SSMS, BIDS, Hive, Hadoop, HDFS, Impala, Sqoop, JIRA.